Comprehensive Tomato (Solanum lycopersicum) Disease, IPM & Agronomy Guide - Page 1

1. Fungal & Oomycete Diseases (Foliar, Fruit & Soil-Borne)

A. Early Blight (Alternaria solani & A. tomatophila)

- Symptoms: Dark brown target-like concentric rings (bull's-eye lesions) with yellow halos on older lower leaves; sunken stem cankers (collar rot) in seedlings; leathery sunken black lesions around the fruit calyx.
- Management: Stake and prune lower leaves (up to 30 cm); apply protectant Mancozeb 75 WP (2.5 g/L) or Chlorothalonil; rotate systemic strobilurins/SDHIs: Azoxystrobin 23 SC (1 ml/L), Boscalid, or Difenoconazole (0.5 ml/L) to prevent FRAC Group 11 resistance.

B. Late Blight (Phytophthora infestans - Oomycete)

- Symptoms: Rapidly expanding, irregular pale-green to dark water-soaked foliar lesions; white downy sporangia on leaf undersides in cool, wet weather (15-22 C, >90% RH); greasy brown-bronze firm rot on green and ripe fruit.
- Management: Forecast via Blightcast models; grow Ph-2/Ph-3 resistant hybrids; prophylactic Mancozeb spray; curative targeted sprays of Cymoxanil + Mancozeb (2.5 g/L), Dimethomorph 50 WP (1 g/L), Mandipropamid 23.4 SC (0.8 ml/L), or Oxathiapiprolin (FRAC 49).

C. Fusarium Wilt (Fusarium oxysporum f. sp. lycopersici - Races 1, 2, 3) & Verticillium Wilt (V. dahliae)

- Symptoms: Unilateral yellowing of lower leaflets (often on one side of petiole/plant), progressive upward wilting during midday heat, stunted growth, and reddish-brown discoloration of xylem vascular tissue when stem is sliced near soil line.
- Management: Plant I-1, I-2, I-3 (Fusarium) and Ve (Verticillium) resistant hybrids; graft scions onto resistant interspecific rootstocks (e.g., S. habrochaites); maintain soil pH 6.5-7.0 using nitrate-N (avoid ammonium-N); drench Trichoderma harzianum (10 g/L).

D. Septoria Leaf Spot (Septoria lycopersici), Gray Mold (Botrytis cinerea), Powdery Mildew & Southern Blight

- Septoria: Numerous small (2-3 mm) circular spots with gray-white centers, dark margins, and tiny black pycnidia. Spray Copper/Azoxystrobin.
- Gray Mold: Ghost spots (pale halos) on fruit and fuzzy gray-brown conidia on pruning wounds in humid greenhouses. Spray Fenhexamid/Fludioxonil.
- Powdery Mildew (Leveillula taurica / Oidium neolycopersici): Talc-like white patches or yellow blotches. Spray Wettable Sulfur (2 g/L) or Myclobutanil.
- Southern Blight / Collar Rot (Sclerotium rolfsii): Rapid wilting with white fan-like mycelial mat and mustard-seed-like brown sclerotia at collar.

2. Major Bacterial Diseases

- Bacterial Wilt (Ralstonia solanacearum): Sudden green wilting without foliar yellowing; pith browning; milky bacterial ooze streams from cut stem suspended in clear water. Control: Bio-fumigation with brassicas; graft onto Hawaii 7996/eggplant rootstock; drench Pseudomonas fluorescens.
- Bacterial Canker (Clavibacter michiganensis subsp. michiganensis): 'Firing' (marginal necrosis) of leaflets, longitudinal stem cankers, and diagnostic 'bird's-eye' spots (dark center with white halo) on fruit. Control: Hot water seed treatment (50 C for 25 min); tool disinfection.
- Bacterial Spot (Xanthomonas spp.) & Speck (Pseudomonas syringae pv. tomato): Greasy water-soaked leaf spots and raised scabby fruit lesions. Control: Spray Copper Hydroxide + Mancozeb tank-mix (fixed copper mobilization), Streptomycin (where permitted), or Acibenzolar-S-methyl.

3. Emerging & High-Impact Viral Diseases & Phytoplasma

- Tomato Brown Rugose Fruit Virus (ToBRFV - Tobamovirus): CRITICAL GLOBAL THREAT. Breaks Tm-2^2 resistance gene. Causes mosaic, fern-leaf narrowing, necrotic peduncles, and wrinkled brown 'rugose' blotches on fruit. Spread mechanically by hands, tools, bumblebees, and seed. Control: Certified GSPP seed; dip tools in 2% Virkon S or 10% trisodium phosphate (TSP); deploy newly bred ToBRFV intermediate-resistant hybrids.
- Tomato Yellow Leaf Curl Virus (TYLCV - Begomovirus): Vectored by Whitefly (Bemisia tabaci). Causes severe stunting, upward cup-like leaf curling, interveinal chlorosis, and flower drop. Control: Grow Ty-1/Ty-2/Ty-3/Ty-6 gene pyramids; 50-mesh insect-proof net houses; vector control.
- Tomato Spotted Wilt Virus (TSWV - Orthotospovirus): Vectored by Western Flower Thrips (Frankliniella occidentalis). Causes bronzing of upper leaves, tip dieback, and striking concentric yellow/orange ringspots on ripe fruit. Control: Sw-5 resistant cultivars; blue sticky traps; Spinetoram.

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4. Key Insect Pests, Vectors & Root-Knot Nematodes (IPM & ETL)

- South American Tomato Pinworm (Tuta absoluta): Devastating invasive pest. Larvae mine leaf mesophyll creating silvery blotch mines with dark frass, and bore pinholes under fruit calyx. ETL: 1-2 mines/plant or 3-5 moths/pheromone trap/day. Control: Mass trapping (25-40 water/delta traps/ha); Nesidiocoris tenuis mirid bug; Trichogramma achaeae; spray Chlorantraniliprole 18.5 SC (0.3 ml/L), Spinetoram, or Cyantraniliprole.
- Tomato Fruit Borer (Helicoverpa armigera): Larvae bore circular holes into developing fruits with half body outside. ETL: 1 larva/plant or 2% damaged fruit. Control: African Marigold (Tagetes erecta) trap crop (1:16 row ratio); HaNPV (250 LE/ha); Emamectin benzoate or Indoxacarb.
- Whiteflies (Bemisia tabaci), Thrips & Aphids: Sap feeders & virus vectors (TYLCV, TSWV, CMV). Control: Yellow/Blue sticky traps (25/ha); reflective silver plastic mulch; spray Flonicamid 50 WG (0.3 g/L), Spiromesifen 22.9 SC (1 ml/L), Afidopyropen, or Azadirachtin 10000 ppm.
- Two-Spotted Spider Mite (Tetranychus urticae) & Russet Mite (Aculops lycopersici): Cause bronze stippling, webbing, and russeted cracked stems. Control: Release predatory mite Phytoseiulus persimilis; spray Abamectin 1.9 EC (0.5 ml/L), Bifenazate, or Fenpyroximate.
- Root-Knot Nematodes (Meloidogyne incognita & M. javanica): Cause root galls, nutrient starvation, wilting, and predispose roots to Fusarium/Ralstonia. Control: Soil solarization; Mi-1.2 resistant varieties (note: Mi-1 breaks down at soil temps >28 C); apply Purpureocillium lilacinum or Fluopyram.

5. Non-Infectious Physiological & Abiotic Disorders

- Blossom End Rot (BER): Sunken, black, leathery necrotic patch at the blossom (distal) end of fruit. Caused by localized Calcium (Ca2+) deficiency in fruit tissue triggered by erratic soil moisture, high salinity, or excessive Ammonium/Potassium antagonism. Control: Maintain uniform drip moisture; apply Calcium Nitrate (Ca(NO3)2) via fertigation; foliar spray 0.5% Calcium Chloride or chelated Ca + Boron directly onto young fruit clusters.
- Fruit Cracking (Radial & Concentric): Caused by rapid water influx after dry spells and wide day/night temp swings. Control: Uniform irrigation + Boron.
- Blotchy Ripening, Yellow Shoulder & Graywall: Green/yellow hard patches on ripe fruit due to Potassium (K) deficiency, high fruit temp (>32 C inhibits lycopene synthesis), or whitefly feeding. Control: Maintain K:N ratio of 2:1 during fruiting; spray 2% Potassium Sulfate (0:0:50).
- Catfacing, Puffiness & Sunscald: Catfacing (puckered scars) occurs from cold temps (<13 C) during flower bud development; Sunscald (white blistered patches) results from over-pruning or defoliation. Control: Avoid excessive pruning on sun-exposed sides; apply kaolin clay particle film.

6. Modern Agronomic Practices, Fertigation & Biosecurity

- Nursery & Grafting: Raise seedlings in sterilized cocopeat protrays inside 50-mesh insect-proof net houses; splice-graft elite scions onto vigorous solanaceous rootstocks (Maxifort, Beaufort, or wild brinjal Solanum torvum) to conquer Bacterial Wilt, Fusarium, Nematodes, and root hypoxia.
- Mulching & Drip Irrigation: Lay 25-30 micron silver-on-black polyethylene mulch on raised beds (30 cm high); silver reflects UV light to repel aphids/whiteflies/thrips, while black suppresses weeds and stops soil-to-leaf spore splash of Early/Late Blight and Septoria.
- Precision Fertigation (Open-Field Hybrid Target: 60-80 t/ha): Total nutrient budget ~180-200 kg N, 100-120 kg P2O5, 220-250 kg K2O/ha. Stage 1 (Establishment, 1-20 DAT): High Phosphorus (19:19:19 + 12:61:0). Stage 2 (Vegetative to Flowering, 21-50 DAT): Balanced NPK + Ca(NO3)2 + Borax (prevents flower drop). Stage 3 (Fruit Set to Ripening, 51-120+ DAT): High Potassium (13:0:45 & 0:0:50) + MgSO4 for Brix and firmness.
- Standard Genetic Resistance Codes (Seed Catalog Key): V (Verticillium), F / Fol: 1,2,3 (Fusarium races), N (Nematodes), T / ToMV (Tomato Mosaic), TSWV (Spotted Wilt), TYLCV (Yellow Leaf Curl), IR: ToBRFV (Rugose Virus), Ph / LB (Late Blight), AB / EB (Early Blight), BW (Bacterial Wilt).